

AI & Education

(The problem with Learning Analytics and the future of AI)

US Public Education (as a system) is...

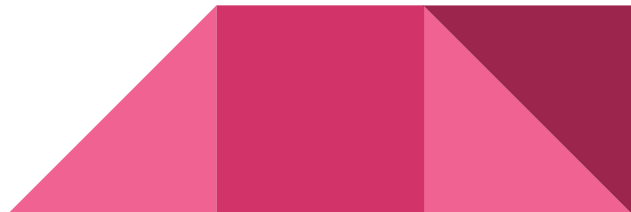
A pipeline to create “productive” citizens

Deficit-based

Designed to rank students and assign potential and value

Teacher-centered learning

Inherently inequitable

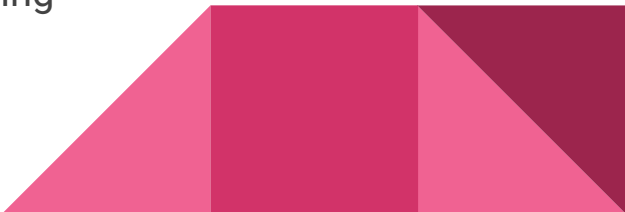


According to Chat GPT...

One way learning analytics is currently being used in US public schools is through the implementation of data-driven decision-making processes. Teachers and administrators can use data analytics to identify areas of student weakness and adjust their teaching methods accordingly. For example, teachers can analyze student performance on assessments and use that data to identify which students need additional support or intervention.

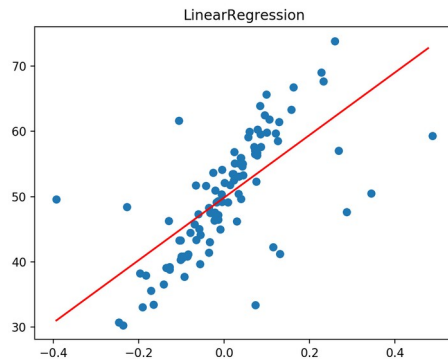
Another way learning analytics is being used in US public schools is through the implementation of learning management systems (LMS). LMSs are software platforms that allow teachers to track student progress, assign and grade assignments, and communicate with students and parents. LMSs also provide data analytics tools that allow teachers and administrators to identify trends and patterns in student learning and adjust their teaching strategies accordingly.

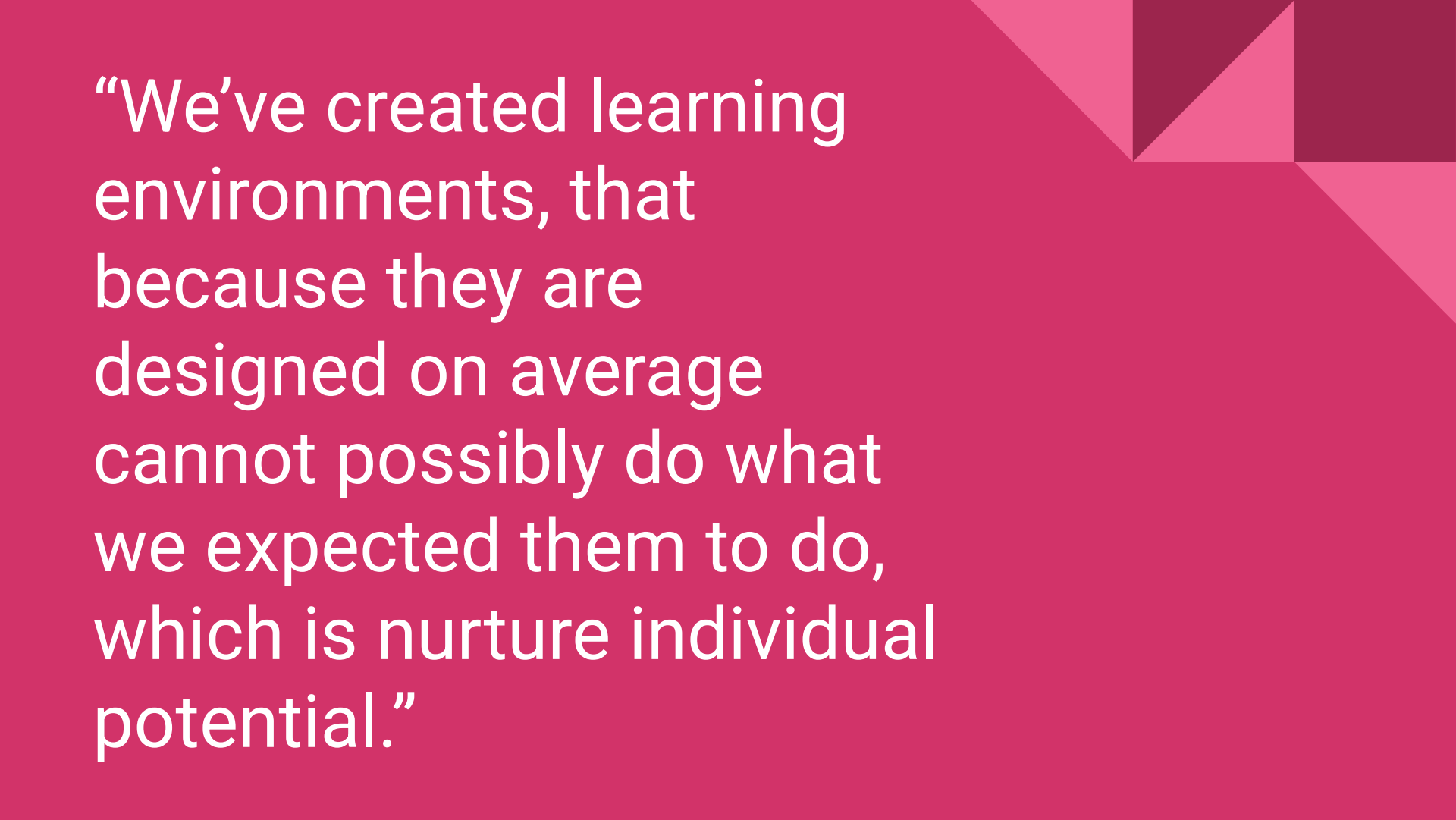
In addition, some US public schools are using predictive analytics to identify students who are at risk of dropping out of school. By analyzing data such as attendance, grades, and behavior, schools can identify students who may need additional support and provide them with targeted interventions to help them succeed.



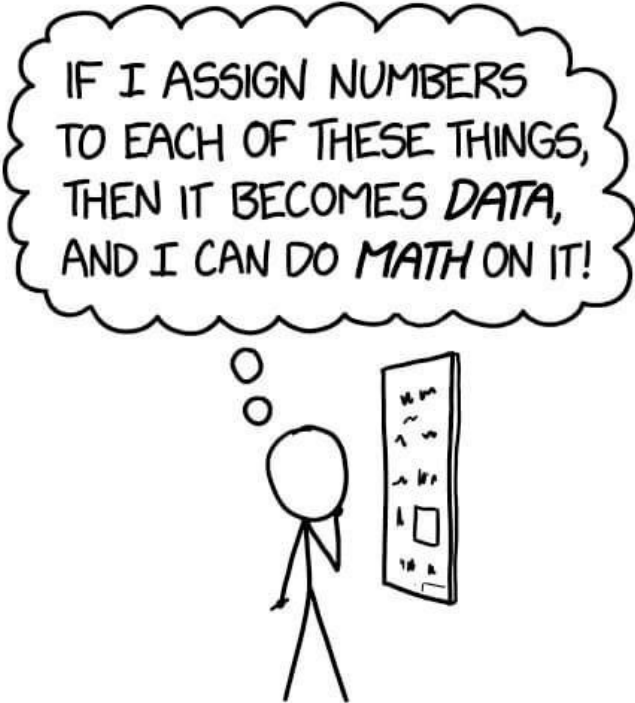
The Problem with Learning Analytics

- LA are using aggregate data to look for patterns from a larger group to apply to a unique, singular individual
- This works “okay” if that individual is “near” or “similar” to the pattern but falls apart if the individual doesn’t fit neatly within the expected pattern





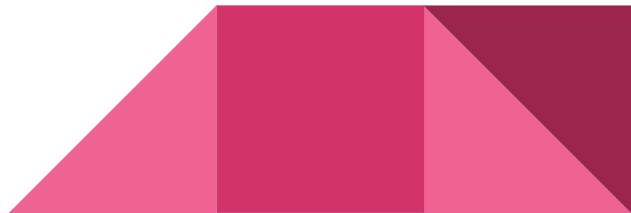
“We’ve created learning environments, that because they are designed on average cannot possibly do what we expected them to do, which is nurture individual potential.”



IF I ASSIGN NUMBERS
TO EACH OF THESE THINGS,
THEN IT BECOMES *DATA*,
AND I CAN DO *MATH* ON IT!

THE SAME BASIC IDEA UNDERLIES
GÖDEL'S INCOMPLETENESS THEOREM
AND ALL BAD DATA SCIENCE.

We can use datalogical systems for efficiency gains and for consistency gains; we can remove that fallible, inconsistent ‘human factor’ in how we make decisions, working more consistently and a million times faster. Which is, you know, fine, sort of. But by definition, a removal of humanity makes a system, well, inhumane! (Keyes, 2019, n.p).





That doesn't mean there is no use for AI in education

Education is highly individualistic

Als can help write individualized curriculum from a more generalized core curriculum

Als can compile and organize evidence of student learning and progress

Als can analyze this body of student work and look for patterns

Note: It is looking for patterns *intra*-student not *inter*-student

Als can help teachers and students create narrative descriptions of their learning and goals as we move away from percentage points and letter grades



Which one tells you more?

Year End Report 2002-2003

Student and parents — redacted

Grade 11 · Birthdate and age — redacted

Days of Attendance 180

Language Arts

———— had a large variety of reading and composition assignments through her work in History of the World. Her reading comprehension skills are excellent, and her composition skills are improving. She did extensive vocabulary work through HOW. We worked on organization and some stylistic techniques using Institute for Excellence in Writing materials.

She participated in Toastmasters for Youth program. She prepared and presented two short speeches and presented several impromptu speeches during the class. Her organization and public speaking skills improved significantly.

■ **Curricula:** Konos History of the World, Year One
Konos History of the World, Year Two
Institute for Excellence in Writing materials

Resources: Elements of Style
Reading list is in her HOW portfolio

Grade 11: 2002-2003

Course	Credits	Grade
Classic Literature/Composition	1.0	A
Algebra 2	1.0	A
World History – Greece and Rome	1.0	A
Chemistry with lab	1.0	A
Latin – Part 3	1.0	B
Logic	0.5	B
Music Appreciation/Piano	0.5	A
Physical Education	0.5	A
Driver's Education	0.5	A
Public Speaking	0.25	B
<i>Cumulative – Year 3</i>	<i>7.25</i>	<i>GPA: 3.76</i>



Design to the Edges

A Pipedream

- Formalized education doesn't start until a child is seven. Everything before that is play-based and experiential
 - This gives us data to create a learner profile and begin to build an individualized curriculum
- AI is used to create a learner profile for each student based on inputs from teachers, the student, and their family
 - This is a living breathing thing that is constantly shifting based on inputs and analysis of that student's work
- Subjects are not siloed into individual classrooms, but rather explored in the context of a student's interests and experiences
- No comparison of students (no class rank, no grades)



Policy Meets Practice

- We don't actually *need* any new learning tools or supports, we have tons at our disposal and we're not using those
- Rather, we need to fundamentally change how we view education in US and our approach to supporting student learning
- We need a national curriculum that is the “edge” for all learners and can be added to as needed for local districts, schools, and students
 - What if, we decided on say the twenty (or be generous and say twenty-five) most important things students needed to know by the time they finished school at a minimum and then used AI to design a curriculum for each student based on their learner profile?



Action Items

- Create a “core curriculum” draft
- PD/Training around narrative assessment
- AI tool to create learning profiles based on multi-modal inputs



References

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